

Lecture 1: Composition and constitution of the Earth

Background(1/3)

- The study of the earth focusses on its layered structure, and the variation in the density and temperature with depth.
- Direct observation of the earth's interior is not possible as the interior becomes hotter with depth.
- Seismological studies and other sources prove that the earth's body comprise of several layers.

Background(2/3)

- Broadly, the earth's interior has been divided into three major layers: the crust; the mantle; the core.
- The crust has a thickness of about 33km.
- The crust is composed of heterogeneous rocks.
- The mantle extends from crust to about 2900km.
- The core extends from the mantle to the centre of the earth.

Background(3/3)

- Seismological studies also provide evidence for:
- Changes in chemical composition of the materials.
- Changes in the density of materials
- Changes in the state of a given material (solid, liquid or viscous)
- Changes in the physical properties of minerals.

Sub-division of the crust

- The **crust** of the earth is sub-divided into Sial and Sima.
- **Sial** – Consists of all types of rocks like igneous, sedimentary and metamorphic which are exposed at the land surface. It is rich in silica and aluminium.
- **Sima** – it is basaltic in composition and the layer is rich in silica and magnesium.

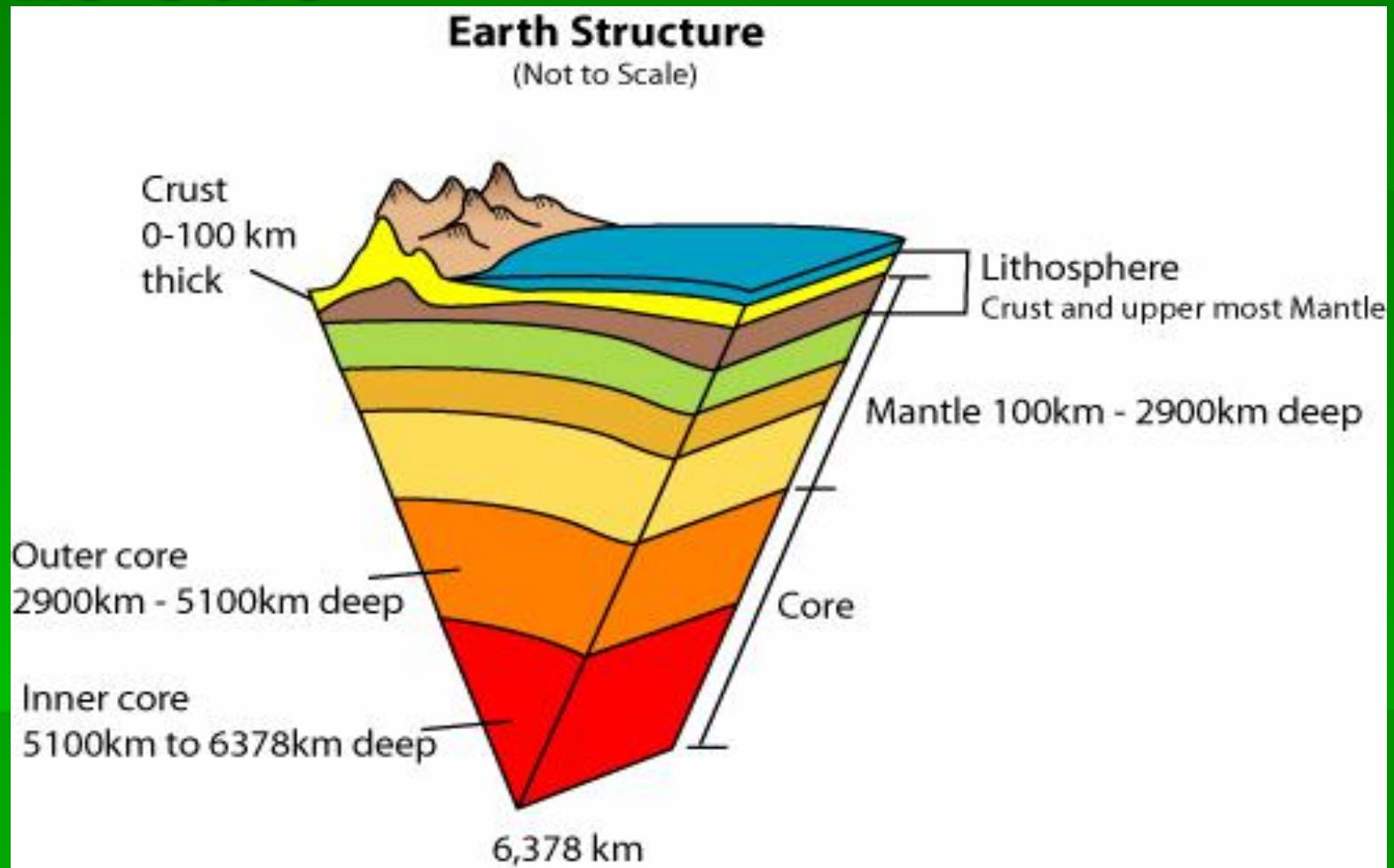
The Mantle

- **Mantle:** its thickness is about 2865km. It is the source-region of most of the earth's internal energy responsible for sea floor spreading orogeny and earthquakes
- The material is olivine-pyroxene complex which exists in solid state.
- Density of the rocks is about 3.36 -3.87.

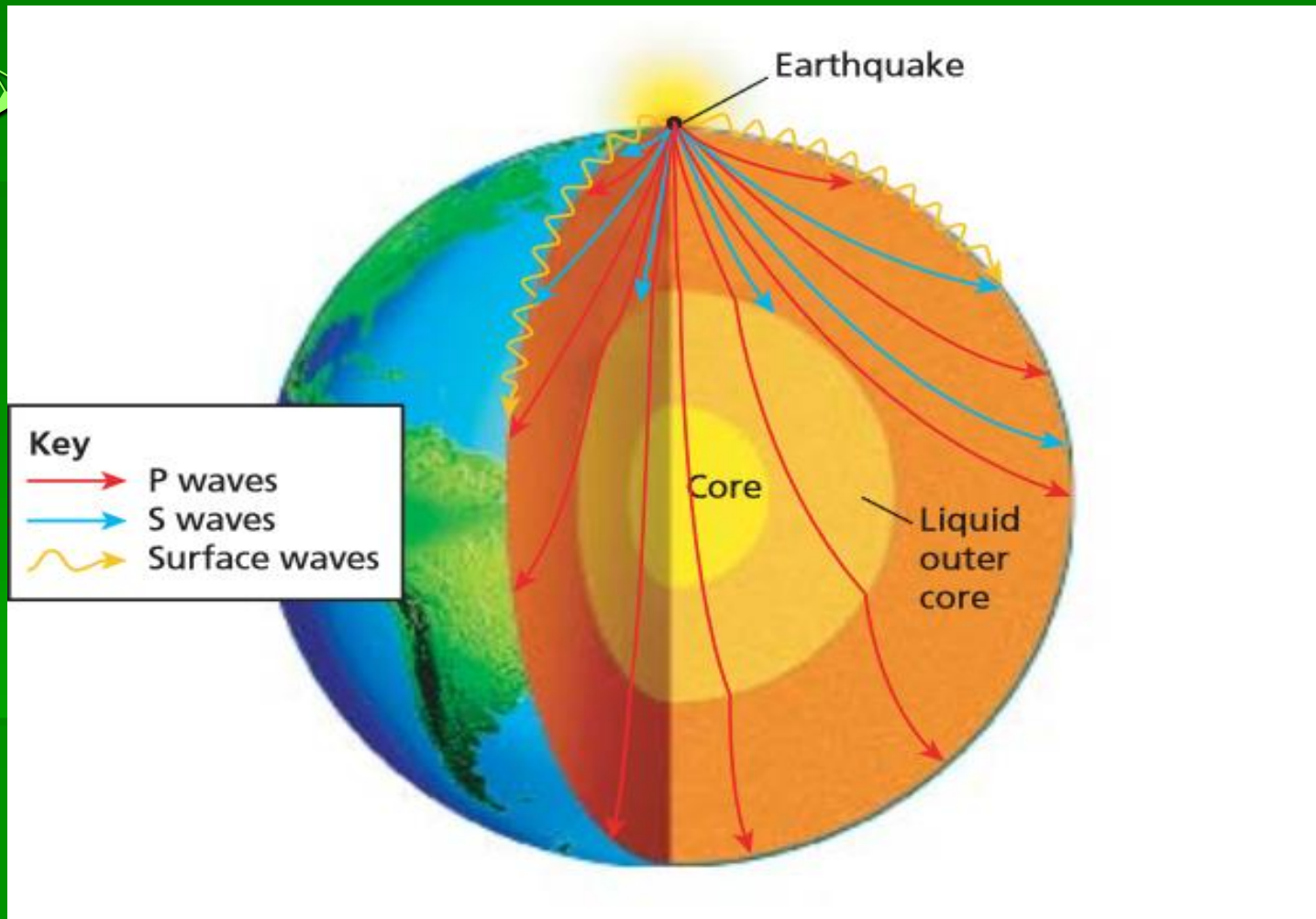
The Core

- The Core – outer core is considered to be in a state of homogeneous fluid and it does not transmit shear waves. The middle core is fluid to semi-fluid. The inner core is believed to contain metallic nickel and iron. It is probably solid with a density of about 18.

The Core

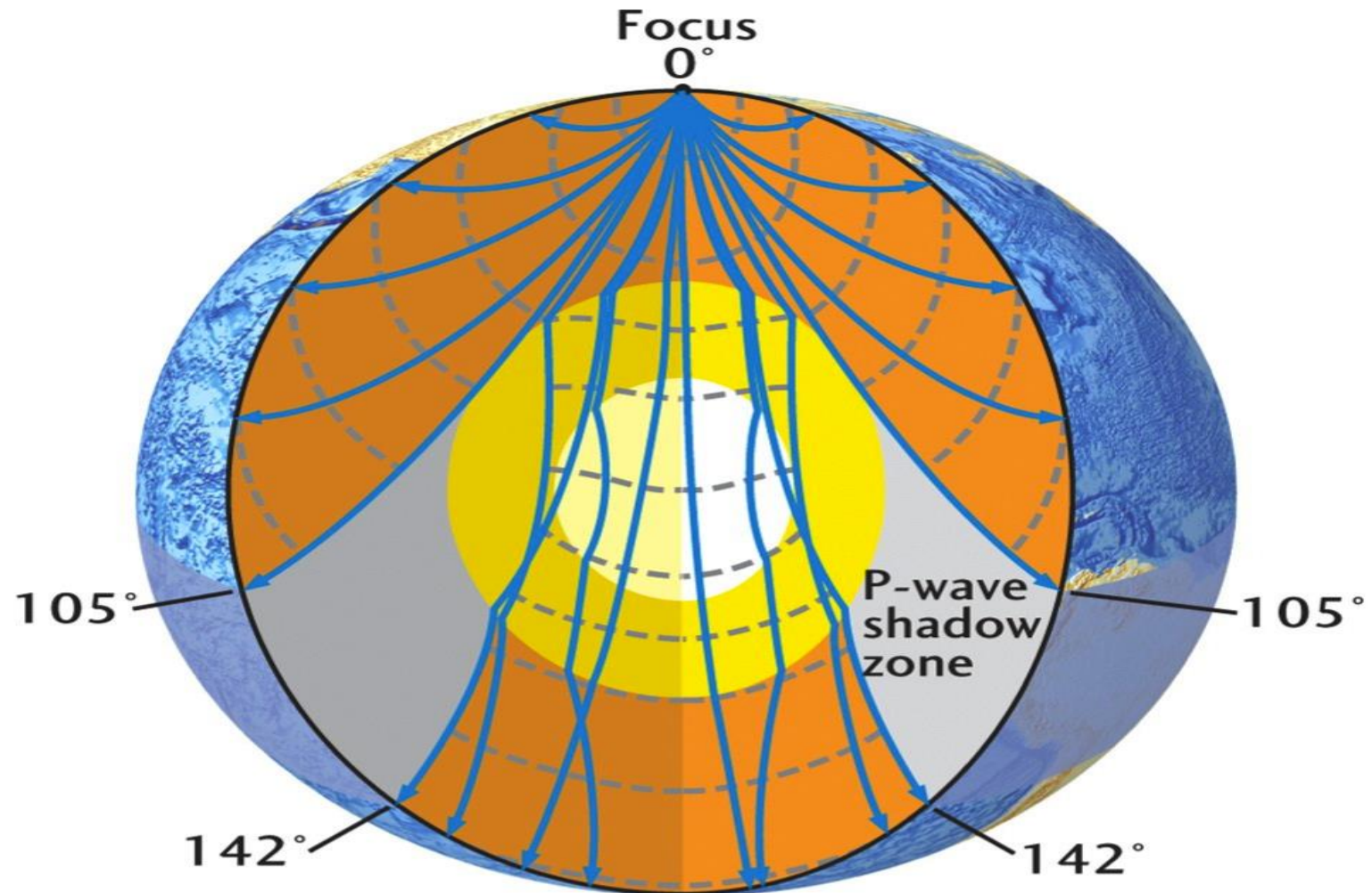


The Core



The Core

The pattern of P-wave paths through Earth's interior



Chemical composition of crustal rocks

- Although over 100 elements occur naturally in minerals, eight elements are so abundant that they make up nearly 99% by weight of all the many thousands of rocks that have been chemically analysed.
- These are oxygen (46.6%), silicon (27.72%), aluminium (8.3%), iron (5.0%), calcium (3.63%), sodium (2.83%), potassium (2.59%), magnesium (2.09%), titanium (0.44%), hydrogen (0.14, phosphorus (0.12%).

Lithosphere and Asthenosphere

- **Lithosphere** – Includes the crust and the uppermost part of the mantle.
- **Asthenosphere** – It is the highly viscous but solid rock portion of the upper mantle wrapping around the core 80 – 120km below the surface.